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CSC 203 — Past Questions & Materials

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Institution: Ambrose Alli University, Ekpoma

Papers in this pack: 1

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Paper 1 of 1 — CSC 203/104

Level 200 • First Semester • PastQuestion

Exam 2019, questions and answers

Computer Programming (CSC 104)

1 of 1

Chapter 1

34 questions

11 fill in the blank

8 true/false

15 multiple choice

Fill in the Blank Questions

1. The _____ is the part of a computer that follows instructions.

[processor]

2. A(n) _____ is a program that translates programs to a simpler language that the computer can execute.

[compiler]

3. The Java interpreter translates a Java program from _____ to machine language.

[byte code]

4. A(n) _____ is an action that a Java object can perform.

[method]

5. Another term for _____ is information hiding.

[encapsulation]

6. A(n) _____ is a set of instructions that tells the basic strategy for solving a problem.

[algorithm]

7. A(n) _____ is a program meant to be sent from one computer to another across the world-wide web.

[applet]

8. A(n) _____ is a piece of information passed to a method.
[argument]

9. A(n) _____ is a digit that can only have the value 1 or 0.
[bit]

10. A(n) _____ is part of a program that has data associated with it and can perform certain actions.
[object]

Optional Graphics Supplement
Questions:

11. The _____ method of an applet is used to draw graphics and is invoked automatically when the applet runs.
[paint]

True/False Questions

1. Main memory refers to the amount of storage space available on a computer's disk drive.
TRUE FALSE
[FALSE]

2. Java is a high-level language.
TRUE FALSE
2 of 2
[TRUE]

3. The Java compiler translates Java programs into machine language programs.
TRUE FALSE
[FALSE]

4. Pseudocode is a mixture of English and a programming language.
TRUE FALSE
[TRUE]

5. The computer will not display an error message when a logic error occurs.

TRUE FALSE

[TRUE]

6. System.out is an example of a method.

TRUE FALSE

[FALSE]

7. In Java, the double equal sign means "make equal to".

TRUE FALSE

[FALSE]

8. Polymorphism is a classification system for organizing classes in Java.

TRUE FALSE

[FALSE]

Multiple-Choice Questions

1. A bit can have _____ different values.

A. 2 B. 10 C. 100 D. 256

[A]

2. Bytes in auxiliary memory are grouped into _____.

A. bits B. files C. directories D. programs

[B]

3. The kind of language that is easiest for people to use is

A. assembly language B. machine language
C. low-level language D. high-level language

[D]

4. Input to a compiler is called

A. a byte-code file B. source code
C. an executable file D. object code

[B]

5. The term _____ refers to a way of organizing classes that share properties.

A. object-oriented B. encapsulation
C. polymorphism D.

inheritance

[D]

6. _____ means that two or more methods can have different names in the same way that an English word can have two or more meanings.

A. object-oriented B.

encapsulation C. polymorphism D.

inheritance

[C]

3 of 3

7. A(n) _____ occurs when a program has a grammatical mistake like missing punctuation.

A. syntax error B. run-time error C. logic error D. hidden error

[A]

8. If the computer displays an error message while executing your program, the program contains

a(n) _____.

A. syntax error B. run-time error C. logic error D. hidden error

[B]

9. Invoking a method means

A. checking the method for errors
B. designing the method
C. asking the method to do its job
D. revising the method

[C]

10. In Java, a library of classes is called

A. a folder B. a package C. a directory D. an application

[B]

11. The _____ is a program that supervises operation of the entire computer.

A. operating system B. compiler C.

interpreter D. virtual machine

[A]

12. A(n) _____ is a unit of memory that contains eight binary digits.

A. bit B. byte C. object D. class

[B]

13. Which of these is not a high-level language?

A. Java B. BASIC C. C++ D. assembly language

[D]

Optional Graphics Supplement

Questions:

14. The point (0, 0) is in the _____ of the screen in the Java coordinate system.

A. upper left-hand corner B. upper right-hand corner
C. lower left-hand corner D. lower right-hand corner
E. center

[A]

15. Which of the following is information passed by parameter(s) to fillOval?

A. the fill color
B. the line color
C. the width of the bounding rectangle
D. the coordinates of the lower-right corner of the bounding rectangle

[C]

Chapter 2

39 questions

15 fill in the blank

13 true/false

13 multiple choice

Fill in the Blank Questions

8 of 8

[FALSE]

2. What is the value of the expression $(x > 3) \parallel (y < 3)$ if x has the value 1 and y has the value 5?

true false

[false]

3. The == operator can be used to see if two strings have the same characters.

TRUE FALSE

[FALSE]

4. A multibranch if-else statement is one kind of nested if-else statement.

TRUE FALSE

[TRUE]

5. The controlling expression of a switch statement must have a value that is an integer or a character.

TRUE FALSE

[TRUE]

6. The body of any while loop will be executed at least once.

TRUE FALSE

[FALSE]

7. for loops and while loops use the same stopping mechanism.

TRUE FALSE

[FALSE]

8. Java uses complete evaluation for Boolean expressions that use & instead of && and | instead of

TRUE FALSE

[TRUE]

9. In ASCII, a capital 'Z' comes before a small 'a'.

TRUE FALSE

[TRUE]

Optional Graphics Supplement
Questions:

10. The drawstring method takes three parameters: a string and two numbers.

TRUE FALSE

[TRUE]

Multiple-Choice Questions

1. What will be the value of x after the following section of code executes:

```
int x = 5;
```

```
if (x > 3)
```

```
    x = x - 2;
```

```
else
```

```
    x = x + 2;
```

A. 1 B. 3 C. 5 D. 7

9 of 9

[B]

2. What will be the value of y after the following section of code executes:

```
int y = 4, z = 3;
```

```
if (y > 4)
```

```
    z = 2;
```

```
    y = y - 1;
```

```
else
```

```
    z = 1;
```

```
    y = z;
```

A. 1 B. 2 C. 3 D. 4

[A]

3. What will be the value of w after the following section of code executes:

```
int w = 4, q = 3;
```

```
if (q > 5)
```

```
    if (w == 7)
```

```
        w = 3;
```

```
    else
```

```
        w = 2;
```

```
    else
```

```
        if (w > 3)
```

```
            w = 1;
```

else

w = 0;

A. 0 B. 1 C. 2 D. 3

[B]

4. What will be the value of b after the following section of code executes:

```
int a = 4, b = 0;
```

```
if (a < 5)
```

```
    b = 4;
```

```
else if (a < 10)
```

```
    b = 3;
```

```
else if (a > 5)
```

```
    b = 2;
```

```
else
```

```
    b = 1;
```

A. 1 B. 2 C. 3 D. 4

[D]

5. What will the following code print?

10 of 10

```
int color = 3;
```

```
switch (color)
```

```
case 1:
```

```
case 2:
```

```
    System.out.print("red ");
```

```
case 3:
```

```
    System.out.print("blue");
```

```
break;
```

```
case 4:
```

```
    System.out.print("purple");
```

```
break;
```

```
default:
```

```
    System.out.print("gray");
```

A. red B. blue C. purple

D. gray E. red blue

[B]

6. What will the following code print?

```
int count = 1;
```

```
while (count <= 3)
```

```
System.out.print(count + " ");  
count++;
```

A. 1 2 3 B. 1 2 C. 3 D. 1 1 1

[A]

7. What will the following code
print?

```
int count;
```

```
for (count = 4; count > 1; count--)
```

```
System.out.print(count + " ");
```

A. 1 2 3 4 B. 1 2 3 C. 4 3 2 D. 4 3 2 1

[C]

8. A(n) _____ is one kind of
branching statement.

A. compound statement B. for
statement

C. break statement D. switch
statement

[D]

9. A(n) _____ goes in the
parentheses following the word if
in an if-else statement.

A. parameter list B. compound
statement

C. nested if statement D. boolean
expression

[D]

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10. What happens when an
expression uses == to compare
two string variables?

A. The value of the expression will
be true if both strings have the
same characters in them.

B. The value of the expression will
be true if both string variables
refer to the same object.

C. The expression will not be
correct syntax and the program
will not compile.

D. A run-time error will occur.

[B]

11. What is the correct way to end a program's execution normally?

- A. `exit()`; B. `exit(0)`;
C. `System.exit()`; D. `System.exit(0)`;

[D]

Optional Graphics Supplement

Questions:

12. How do you set the color of an oval?

- A. By setting the `drawColor` field of the Graphics class.
B. By calling the `setColor` method of the Graphics class.
C. By calling the `set` method of the Color class.
D. By passing a Color object as a parameter to the `drawOval` method.

[B]

Chapter 4

34 questions

13 fill in the blank

12 true/false

9 multiple choice

Fill in the Blank Questions

1. A(n) _____ is an action that an object can take and is specified in the class definition.

[method]

2. Compiling a file called `Game.java` will produce a file called _____.

[Game.class]

3. A(n) _____ is a data item that belongs to an object.

[instance variable]

4. _____ is the word used to say that there are no restrictions on the use of a method or data item.

[public]

5. The reserved word _____ can be used as a name for the calling object.

[this]

6. Names of mutator methods usually begin with _____.

[set]

7. The interface of a class consists of comments, _____, and defined constants.

[headings of public methods]

8. To create an object of a class type, the word _____ is used.

[new]

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9. One kind of method returns a value. The other kind of method is called a(n) _____.

[void-method]

10. A _____ is the definition of a kind of object.

[class]

11. The _____ of a method definition is enclosed between braces.

[body]

12. A variable declared inside a method is called a _____ variable.

[local]

Optional Graphics Supplement Questions:

13. The _____ method of an applet is called automatically when the applet is starting up.

[init]

True/False Questions

1. All objects of the same class have the same data values.

TRUE FALSE

[FALSE]

2. When a method call is executed, values from the method definition are substituted for the arguments in the method call.

TRUE FALSE

[FALSE]

3. Parameters of a primitive type are passed to methods using the call-by-value mechanism.

TRUE FALSE

[TRUE]

4. Encapsulation is the process of defining a class that has no public methods.

TRUE FALSE

[FALSE]

5. Private methods are part of the interface of a class.

TRUE FALSE

[FALSE]

6. A variable of class type does not store an object of that class.

TRUE FALSE

[TRUE]

7. An equals method should return a boolean value.

TRUE FALSE

[TRUE]

8. A method can change the value of a variable of a primitive type that is an argument to the method.

TRUE FALSE

[FALSE]

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9. A method can change the value of instance variables of an argument that is of a class type.

TRUE FALSE

[TRUE]

10. A Java method that does not

return a value can have any
number of return statements.

TRUE FALSE

[TRUE]

Optional Graphics Supplement

Questions:

11. drawString is a method of the
Graphics class.

TRUE FALSE

[FALSE]

12. An applet can be successfully
compiled even if it has no init
method and no paint method.

TRUE FALSE

[TRUE]

Multiple-Choice Questions

1. A method is invoked by

A. writing the name of the method
followed by a colon and the name
of the calling object

B. writing the name of the calling
object followed by a dot and a list
of values in parentheses

C. listing the name of the calling
object and the name of the method
inside parentheses

D. writing the name of the calling
object, followed by a dot, followed
by the name of the method
and a list of values in parentheses

[D]

2. What can you tell about the
method surpriseMe in the code
below?

```
int something;
```

```
something = surpriseMe();
```

A. It has no parameters but has a
return value.

B. It has one parameter but no
return value.

C. It has no parameters and it has

no return value.

D. It has one return value and an unknown number of parameters.

E. It has one parameter and an unknown number of return values.

[A]

3. A data item that can only be used within a method is called

A. a local variable B. an instance variable

C. a global variable D. a private variable

[A]

4. Normally all instance variables are given the qualifier

A. public B. protected C. private D. static

[C]

5. One benefit of encapsulation is

A. the implementation of the class will be smaller

B. the implementation can be changed without changing programs that use the class

C. the interface of the class will be smaller

D. the interface can be changed without changing programs that use the class

[B]

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6. When comparing two objects to see if they have the same contents, one should

A. use the = operator

B. use the == operator

C. define an equals method in the class and use it

D. use the equals method to see if they have the same variable name

[C]

7. A _____ method does not return a value.

A. null B. static C. void D. recursive

[C]

Optional Graphics Supplement

Questions:

8. Where does the Graphics object of an applet come from?

A. It is usually created in the init method and stored as an instance variable of the applet.

B. It is inherited from the JApplet class.

C. It is a local variable of the applet's init method.

D. It is created automatically and passed in to the applet's paint method as a parameter.

[D]

9. Which of the following is a method of the Graphics class?

A. drawRectangle B. circleFill C. drawOval D. paint

[C]

Chapter 5

28 questions

10 fill in the blank

8 true/false

10 multiple choice

Fill in the Blank Questions

1. A _____ method is a method that can be invoked using the class name instead of an object name.

[static]

2. The method used to convert from Character to char is _____.

[charValue]

3. The method used to convert from String to Float is _____.

[parseFloat]

4. A(n) _____ is a simplified version of a method that is easily verified and is used for testing.

[stub]

5. In the technique of _____ design, the problem of designing a method is divided into subproblems, which are then solved using the same technique.

[top-down]

6. A(n) _____ is a special kind of method used to initialize objects.

[constructor]

7. A(n) _____ is a collection of related classes that serves as a class library.

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[package]

8. The _____ class provides the methods pow and round, among others.

[Math]

9. Integer and Double are _____ classes.

[wrapper]

10. A constructor with no parameters is called a _____ constructor.

[default]

True/False Questions

1. The object and dot can be omitted in a non-static method call if the calling object is the this parameter.

TRUE FALSE

[TRUE]

2. Instance variables cannot be referenced in the definition of any static method unless an object (other than this) and dot precede

the variable name.

TRUE FALSE

[TRUE]

3. Java will use automatic type conversion before it will use overloading.

TRUE FALSE

[FALSE]

4. It is possible to have two methods in the same class that have the same name, the same number and types of arguments, but different return types.

TRUE FALSE

[FALSE]

5. Constructors have return type void.

TRUE FALSE

[FALSE]

6. A class may have more than one constructor.

TRUE FALSE

[TRUE]

7. If you add at least one constructor to a class, no constructors will be automatically created for the class.

TRUE FALSE

[TRUE]

Optional Graphics Supplement

Questions:

8. It is possible to make a component in an applet invisible after the applet has started executing.

TRUE FALSE

[TRUE]

Multiple-Choice Questions

1. The method _____ in the Math class returns the nearest

whole number that is equal to or
16 of 16

less than its argument.

A. abs B. min C. round D. floor

[D]

2. Which of these words is the
name of a wrapper class?

A. int B. float C. Double D. String

[C]

3. The term overloaded refers to

A. an error where more than one
method matches a method call

B. two or more methods in the
same class that have the same
name

C. method calls where there are
two many actual parameters

D. two or more methods in
different classes that have the
same name

[B]

4. Java will automatically define a
constructor

A. if the programmer does not
define a default constructor for a
class

B. if the programmer does not
define any constructors for a class

C. if the program refers to a
constructor with no parameters

D. for every class defined in a
program

[B]

5. A private instance variable

A. can never be modified by a
method outside the class

B. can be modified by a method
outside the class if it is returned
by a method

C. can be modified by a method
outside the class if it is a class

type, using a mutator method
(assuming one exits)

D. can be modified by using an
object name followed by a dot and
then the name of the
instance variable

[C]

6. The name of a constructor

A. can be any legal identifier

B. must include the name of the
package

C. will always be the word new

D. will always be the same as the
name of the class

[D]

7. _____ is a special constant
that is not an object but can be
used to give a value to any
variable of class type.

A. void B. this C. null D. false

[C]

8. A(n) _____ allows you to
have a class object that
corresponds to a value of a
primitive type.

A. wrapper class B. type cast C.
constructor D. static variable

[A]

Optional Graphics Supplement

Questions:

9. When the user clicks a button,
the event will be handled by an
object of type _____.

A. ActionListener B. EventHandler
C. ButtonListener D.

ActionHandler

[A]

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10. In Java, an icon is _____ that
can be added to a component.

A. a menu B. a button C. a sound D.

a picture

[D]

Chapter 6

28 questions

8 fill in the blank

11 true/false

9 multiple choice

Fill in the Blank Questions

1. The type of elements in an array is called the _____.

[base type]

2. Every array object has an instance variable called _____ that tells how many elements are in the array.

[length]

3. An array can be initialized by enclosing a list of values in _____.

[braces]

4. When a program looks at array elements in order from first to last to find a value, it is using the _____ algorithm.

[sequential search]

5. An array that uses more than one index is called a(n) _____ array.

[multidimensional]

6. The parameter to main is a(n) _____.

[array of Strings]

7. A two-dimensional array is implemented as a(n) _____.

[array of arrays]

Optional Graphics Supplement Questions:

8. The _____ method is the same as the drawPolygon method, except that it does not draw a

line from the last point back to the first point.

[drawPolyline]

True/False Questions

1. All data stored in an array must be the same type.

TRUE FALSE

[TRUE]

2. The index to use with an array can be an expression whose value is computed when the program runs.

TRUE FALSE

[TRUE]

3. If one element from an array is passed as a parameter to a method, then the called method can

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modify any element in the array.

TRUE FALSE

[FALSE]

4. Every array has a method called equals that can be used to compare it to other arrays.

TRUE FALSE

[FALSE]

5. An accessor method that returns an array should return a copy of an instance variable array rather than the instance variable itself.

TRUE FALSE

[TRUE]

6. In a two-dimensional array, all of the rows must have the same number of elements.

TRUE FALSE

[FALSE]

7. The size of an array in Java is determined when the program is

compiled.

TRUE FALSE

[FALSE]

8. You can change the size of an array by assigning a new value to the length instance variable.

TRUE FALSE

[FALSE]

9. The first element in an array is accessed using zero as an index.

TRUE FALSE

[TRUE]

10. Arrays in Java are objects.

TRUE FALSE

[TRUE]

11. Selection sort is not one of the easier to understand sorting algorithms, but it is one of the most efficient.

TRUE FALSE

[FALSE]

Multiple-Choice Questions

1. The lowest index that can be used with an array

A. is zero

B. is one

C. is determined by the array declaration

D. changes as the program runs

[A]

2. The highest index that can be used with an array

A. is the number of elements in the array

B. is the number of elements in the array minus one

C. is determined by the array declaration

D. changes as the program runs

[B]

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3. What will happen if a program uses an array index that is out of bounds?

A. The compiler will give an error message and will not compile the program.

B. The compiler will give a warning but will still compile the program.

C. The program will compile but Java will give an error message when the program runs.

D. The program will compile and run with no error messages but might give incorrect results.

[C]

4. Which statement best describes an array of 10 elements where the elements are of class String?

A. 10 String objects will be created when the array is created.

B. 10 String objects must be created before the array is created.

C. The array will be created with the elements equal to null.

D. Elements of an array cannot be a class type.

[C]

5. Selection sort works by repeatedly

A. using the position of elements to split the list into two sublists of equal size

B. finding the smallest element in the remainder of the list and moving it forward

C. comparing adjacent elements in the list and swapping them if they are out of order

D. using a threshold value to split the list into two sublists where one sublist's elements are greater than the threshold and the other sublist's elements are less than the threshold

[B]

6. Which of the following is correct Java code?

A. `int[10] list;`

B. `int list[10];`

C. `int list = new int[10];`

D. `int[] list = new int[10];`

[D]

7. The main method of a program has

A. no parameters

B. one String parameter

C. one parameter which is an array of String values

D. an array of integers as the first parameter and an array of strings as the second parameter

[C]

Optional Graphics Supplement Questions:

8. What is the main difference between a text field and a text area?

A. A text area can display more than one line and a text field can only display one line of text.

B. A text field can be changed by the user and a text area cannot.

C. A text area has one or more text fields inside it.

D. A text area has a `setText` method but a text field does not.

[A]

9. How are the points of a polygon specified?

- A. By repeated calls to the drawPolygon method, with each call specifying one x coordinate and one y coordinate.
- B. By passing an array of numbers that contains alternating x and y coordinates.
- C. By passing two arrays of numbers, one containing x coordinates and the other containing y coordinates.

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- D. By passing in a two-dimensional array of numbers that contains both x and y coordinates.

[C]

Chapter 7

29 questions

12 fill in the blank

12 true/false

5 multiple choice

Fill in the Blank Questions

1. A(n) _____ is defined by adding instance variables and methods to an existing class.

[derived class or child class]

2. Inheritance can be used to create a new class by adding to an existing _____ class.

[base or parent]

3. When a class has a method that has the same name and the same number and types of parameters as an inherited method, then the method _____ the inherited method.

[overrides]

4. The constructor of a derived class can access the constructor of

its base class by using the
reserved word _____.

[super]

5. The reserved word _____ can
be used as a name for another
constructor in the same class.

[this]

6. The reserved word _____
between two class names at the
beginning of a class definition
indicates that the first class
inherits from the second.

[extends]

7. In Java, every class is a
descendant of the class _____.

[Object]

8. _____ binding is when the
meaning of a method invocation is
not bound to it until the
program is run.

[dynamic or late]

9. _____ means "many forms".
In programming, it allows
different objects to use different
method actions for the same
method name.

[polymorphism]

10. A(n) _____ method is one
which has no body and is meant to
be overridden in every derived
class.

[abstract]

11. Adding the _____ modifier
to a method heading prevents
derived classes from overriding
the
method.

[final]

12. A(n) _____ specifies
headings for methods that must be
defined by an implementing class.

[interface]

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True/False Questions

1. Instance variables that are private in a base class can be accessed in the definition of a method in a derived class.

TRUE FALSE

[FALSE]

2. Private methods are not inherited.

TRUE FALSE

[TRUE]

3. If a constructor includes a call to the constructor of the base class, then the call must be the first action in the constructor.

TRUE FALSE

[TRUE]

4. An object of a derived class has the type of the derived class and also the type of every ancestor class.

TRUE FALSE

[TRUE]

5. A method in a derived class may not re-use any method name used in an ancestor class.

TRUE FALSE

[FALSE]

6. If class Able inherits from class Baker, and class Baker inherits from class Charlie, then class Able inherits from class Charlie.

TRUE FALSE

[TRUE]

7. The keyword this can be used as a method name that refers to a constructor.

TRUE FALSE

[TRUE]

8. The expression
super.super.foo() is legal as long
as the foo method is defined in the
appropriate
class.

TRUE FALSE

[FALSE]

9. You can assign an object of an
ancestor class to a variable of a
derived class type.

TRUE FALSE

[FALSE]

Optional Graphics Supplement

Questions:

10. Every class derived from
JFrame has a paint method.

TRUE FALSE

[TRUE]

11. The close-window button of a
JFrame GUI generates an event
which is handled by an action
listener.

TRUE FALSE

[FALSE]

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12. The setSize method is
commonly used to change the size
of an applet.

TRUE FALSE

[FALSE]

Multiple-Choice Questions

1. The expression super.foo() in a
method

- A. is a call to a method named
super.foo in the same class
- B. is a call to the foo method in the
base class
- C. is a call to the constructor of the
class named foo

D. is a syntax error

[B]

2. Suppose that class Able inherits from class Baker, with a1 and b1 referring to objects of class Able and Baker respectively.

Which statement best describes how objects of classes Able and Baker could be assigned to variables?

A. an object of class Able could be assigned to b1, and an object of class Baker could be

assigned to a1

B. an object of class Able could be assigned to b1, but an object of class Baker could not be assigned to a1

C. an object of class Baker could be assigned to a1, but an object of class Able could not be assigned to b1

D. an object of class Able could not be assigned to b1, and an object of class Baker could not be assigned to a1

[B]

3. What determines whether a method definition in a derived class or an ancestor class will be used?

A. the return type of the method

B. the type of the variable that names the object

C. the object's place in the inheritance chain

D. the derived class's descendants

[C]

4. An abstract class

A. is a class which cannot be instantiated

- B. is a class which has no methods
- C. is a class which has only abstract methods
- D. is a class which has only overridden methods

[A]

5. Which statement best describes overriding a method?

- A. Methods in a base class and derived class have the same name but different visibility modifiers.
- B. Methods in a base class and derived class have the same name and have the same number and types of parameters.
- C. Methods in a base class and derived class have the same name but have different return types.
- D. Methods in a base class and derived class have the same name but different number or types of parameters.

[B]

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Chapter 8

25 questions

7 fill in the blank

9 true/false

9 multiple choice

Fill in the Blank Questions

1. The part of the code that tells what the program will do in normal circumstances is called the _____.

[try-block]

2. The part of the code that tells what to do when an exception occurs is called the _____.

[catch-block]

3. An exception is a(n) _____.

[object]

4. Exception-handling code can get more information about the exception by using the _____ method.

[getMessage]

5. A method that includes a throw statement but does not have a corresponding catch must include a(n) _____ in its heading.

[throws-clause]

6. Statements in a(n) _____ block will be executed regardless of whether or not an exception is thrown.

[finally]

7. _____ are often the only methods in an exception class, other than inherited methods.

[constructors]

True/False Questions

1. A throw statement is only allowed inside a catch-block.

TRUE FALSE

[FALSE]

2. A program is not allowed to have more throw statements than catch-blocks.

TRUE FALSE

[FALSE]

3. Programmer-defined exception classes must be derived from a previously defined exception class.

TRUE FALSE

[TRUE]

4.

ArrayIndexOutOfBoundsException is an exception that does not need to be caught or declared in

the method heading.

TRUE FALSE

[TRUE]

5. catch-blocks should be in order from most general to most specific.

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TRUE FALSE

[FALSE]

6. catch-blocks may have any number of parameters.

TRUE FALSE

[FALSE]

7. Every exception class is an ancestor of the class Exception.

TRUE FALSE

[FALSE]

8. Any method that includes a throw statement must also include a catch-block.

TRUE FALSE

[FALSE]

9. Exceptions that are descendants of the class RuntimeException do not have to be caught or declared.

TRUE FALSE

[TRUE]

Multiple-Choice Questions

1. A catch-block

- A. is a special method used in exception handling
- B. allows the user to specify what will happen next in the program
- C. is not allowed in the same method as a try-block
- D. contains code that is executed when an exception occurs

[D]

2. The purpose of the exception-handling mechanism in Java can

best be described as

- A. to simplify the code for normal circumstances
- B. to provide an alternative for if-else statements
- C. to reduce running time of programs
- D. to reduce debugging time of programs

[A]

3. When an object is thrown

- A. it is deleted and can no longer be used
- B. execution resumes with the next executable statement
- C. control passes to a catch-block
- D. the object is unavailable until the catch-block finishes execution

[C]

4. When a catch-block finishes execution

- A. the program continues in the try-block that was previously executing
- B. the program returns to the beginning of the try-block to re-execute it.
- C. the program continues with the next statement after the try-block and catch-blocks
- D. the program stops execution

[C]

5. A programmer-defined exception class

- A. can be derived from any existing class
- B. should provide two or more constructors that set up the message for the exception

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- C. should not have more than one

instance variable

D. should contain at least one int variable to store the exception code number

[B]

6. Overuse of exceptions

A. is not an issue because exceptions may be used freely

B. is not an issue because the compiler restricts programmer-defined exceptions

C. is a problem because it causes an excessive number of class definitions

D. is a problem because it results in unrestricted flow of control

[D]

7. catch-blocks are located

A. at the beginning of the method

B. at the end of the method

C. before a try-block

D. after a try-block

[D]

8. The parameter to the Exception class constructor

A. is a String that will be the message for the exception

B. is an int that will be the exception code number

C. tells the statement where execution should resume

D. is a boolean telling whether or not the program should terminate

[A]

9. What happens if the catch-block of a more general exception comes before the catch-block of a more specific exception?

A. Both catch-blocks will execute when the more general exception occurs.

B. Both catch-blocks will execute when the more specific exception occurs.

C. The more general catch-block will never execute.

D. The more specific catch-block will never execute.

[D]

Chapter 9

28 questions

11 fill in the blank

10 true/false

7 multiple choice

Fill in the Blank Questions

1. A(n) _____ is an object that takes data from a source or delivers data to its destination.

[stream]

2. Files that are considered to be a string of characters and can easily be opened with an editor are called _____ files.

[text]

3. The preferred class for output to a text file is _____.

[PrintWriter]

4. _____ to a file means to add data to the end of the file.

[appending]

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5. The method _____ in the BufferedReader class can be used to read a line of input from a file.

[readLine]

6. The class _____ can be used to break a string up into individual words.

[StringTokenizer]

7. All output streams have a method called _____ that writes

all buffered data to the file.

[flush]

8. The class _____ provides a method to find out if a given file already exists.

[File]

9. When a program tries to read beyond the end of a file, the read method returns the value _____.

[-1]

10. The print method of PrintWriter will automatically call the _____ method of objects passed to it as parameters.

[toString]

11. A(n) _____ name gives the name of a file and also tells what directory (or folder) the file is in.

[path]

True/False Questions

1. A disadvantage of binary files is that are less efficient to process than other files.

TRUE FALSE

[FALSE]

2. A file variable declared inside a try-block will not be accessible outside of the try-block.

TRUE FALSE

[TRUE]

3. Careless use of the FileOutputStream constructor can result in lost files.

TRUE FALSE

[TRUE]

4. When a program will be appending to a file rather than creating a new file, the string

"append"

should be added as a second parameter to the constructor for `FileOutputStream`.

TRUE FALSE

[FALSE]

5. The preferred way to open a text file for input is to pass the file name to the constructor for `BufferedReader`.

TRUE FALSE

[FALSE]

6. The way to find out if a tokenizer has more tokens is to use the boolean instance variable `more`.

TRUE FALSE

[FALSE]

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7. The `readLine` method in `BufferedReader` returns the value `null` if there is no more data to read from the file.

TRUE FALSE

[TRUE]

8. `ObjectInputStream` methods that try to read past the end of a file will cause an `EOFException`.

TRUE FALSE

[TRUE]

9. If data in a stream flows out of a file then the stream is an output stream.

TRUE FALSE

[FALSE]

10. Your program should call the `close` method when it is finished with a file because otherwise the next program that needs to use the file will not be able to open it.

TRUE FALSE

[FALSE]

Multiple-Choice Questions

1. Every program that does file I/O

- A. must include the line `include io.java` somewhere in the program
- B. can do input from a file or output to a file but not both
- C. must be designed specifically for the operating system in which it will run
- D. must include the line `import java.io.*` near the beginning

[D]

2. Opening a text file for writing

- A. requires use of the `PrintWriter` constructor
- B. requires use of the `FileOutputStream` constructor
- C. requires use of the `PrintWriter` constructor and the `FileOutputStream` constructor
- D. does not require use of any constructors

[C]

3. If a new output text file cannot be created

- A. a `FileNotFoundException` will be thrown
- B. a `FileNotCreatedException` will be thrown
- C. the program will stop executing with an error message
- D. the program will continue executing and might produce incorrect results

[A]

4. If an open file is not closed when the program finishes executing

- A. a `FileNotClosed` exception will

be thrown

B. the contents of the file will be lost

C. Java will close the file but it is better to explicitly close the file

D. no other program will be able to open the file

[C]

5. The classes used to open a binary file for writing are

A. PrintWriter and
FileOutputStream

B. ObjectOutputStream and
FileOutputStream

C. PrintWriter and
ObjectOutputStream

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D. BufferedWriter and
ObjectOutputStream

[B]

6. A good programming practice for I/O in Java is

A. have objects do their own I/O

B. design a single class to do all I/O for the program

C. avoid file I/O since it is inefficient

D. never append to a file

[A]

7. If the second parameter to the FileOutputStream constructor is _____, a program will be able to append to a file.

A. the boolean value true

B. the character 'a' or 'A'

C. the character '+'

D. the string "append"

[A]

Chapter 10

26 questions

12 fill in the blank

10 true/false

8 multiple choice

Fill in the Blank Questions

1. Vectors can be thought of as _____ that can grow and shrink.

[arrays]

2. The vector method that takes an index as a parameter and returns an element of the vector is _____.

[elementAt]

3. The vector method that tells how many elements are currently in the vector is _____.

[size]

4. Usually the _____ node is the only node in the list for which there is a named variable.

[head]

5. A(n) _____ is an object that allows you to step through a list and perform some action for each element of the list.

[iterator]

6. To make a linked list that can contain different kinds of elements, the data instance variable should be defined as _____.

[Object]

7. A(n) _____ is a linked list where a program can easily move backward in the list as well as forward.

[doubly-linked list]

8. The vector method _____ can be used to reduce the amount of memory used by the vector to the minimum necessary.

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[trimToSize]

9. Each node in a linked list has some data and a(n) _____ to another node.

[link or reference]

10. The process of recycling the memory used by deleted nodes is called automatic _____.

[garbage collection]

11. A class defined within another class is called a(n) _____.

[inner class]

12. The _____ of a vector is the type of elements that can be added to the vector.

[base type]

True/False Questions

1. A program that uses vectors must include the line: `import java.vector.*;`

TRUE FALSE

[FALSE]

2. If the current size of a vector is 10, the method call `v.setElementAt(newEl, 10)` will add newEl as

the 11

th

element of the vector.

TRUE FALSE

[FALSE]

3. Any vector that is not parameterized (has no type parameter) will have a base type of Object.

TRUE FALSE

[TRUE]

4. A tree is a linked data structure where links form loops among the elements.

TRUE FALSE

[FALSE]

5. A vector created with the default constructor will have 10 elements and will add 10 elements whenever it needs to increase its capacity.

TRUE FALSE

[FALSE]

6. A vector can grow and shrink during execution of a program.

TRUE FALSE

[TRUE]

7. The square bracket notation used with arrays can also be used with vectors.

TRUE FALSE

[FALSE]

8. The base type of a vector is specified in angular brackets after the class name, Vector.

TRUE FALSE

[TRUE]

9. A primitive type cannot be the base type of a vector.

TRUE FALSE

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[TRUE]

10. The Vector method `elementAt` always returns a result of type `Object`.

TRUE FALSE

[FALSE]

Multiple-Choice Questions

1. One disadvantage of vectors is

- A. they have a limited capacity
- B. they can only contain primitive types
- C. they can only contain objects
- D. they often cause exceptions

[C]

2. A number passed as an

argument to the Vector

constructor

- A. causes a compiler error
- B. indicates the type of elements that will be stored in the vector
- C. determines the initial capacity of the vector
- D. makes the number the first element of the vector

[C]

3. A program can make a copy of a vector by

- A. using the assignment operator
- B. using the clone method of Vector
- C. using the copy method of Vector
- D. writing special-purpose code

[B]

4. Links in a linked list are usually

- A. references B. integers C. arrays
- D. vectors

[A]

5. In the textbook's implementation of a linked list, the statement used to go from one node to the

next was a statement like:

- A. `position.next();`
- B. `position++;`
- C. `position.addElement(1);`
- D. `position = position.getLink();`

[D]

6. What happens to nodes that are deleted from a linked list?

- A. The memory they used is reclaimed for other uses automatically.
- B. The memory they use is reclaimed when the method `System.garbageCollect` is called.
- C. The memory they use is not

reclaimed until the head node is deleted.

D. The memory they use is not available for other uses until the program terminates.

[A]

7. Where is the easiest place to add new nodes to a linked list?

A. At the beginning of the list.

B. At the end of the list.

C. Wherever the current position is.

D. Wherever there is an empty position.

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[A]

8. The elementAt method

A. has two parameters

B. only returns values of primitive types

C. is typically used with a type cast

D. returns null if the requested element does not exist

[C]

Chapter 11

25 questions

5 fill in the blank

11 true/false

9 multiple choice

Fill in the Blank Questions

1. When a method definition contains an invocation of itself, the method is said to be _____.

[recursive]

2. A _____ is a case with no recursive calls.

[stopping case or base case]

3. If every case of a recursive method includes calls to itself, the result will be _____.

[infinite recursion]

4. A rewritten recursive method that uses a loop instead of recursion is called a(n) _____ version of the method.

[iterative]

5. Binary search cannot be used with an array that is not _____.

[sorted]

True/False Questions

1. The main reason for using recursion is to make a program easier to understand.

TRUE FALSE

[TRUE]

2. A recursive method needs to have an if-else or some other branching statement that leads to different cases.

TRUE FALSE

[TRUE]

3. A recursive method should only contain one call to itself.

TRUE FALSE

[FALSE]

4. A recursive method must contain at least one case where it does not call itself.

TRUE FALSE

[TRUE]

5. The error that results from having too many stopping cases is called stack overflow.

TRUE FALSE

Saved